

HACK THE BOX SHERLOCK - DFIR CAMPFIRE-1

```
(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1]
$ ls
campfire-1.zip  Triage

(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1]
$ cd Triage

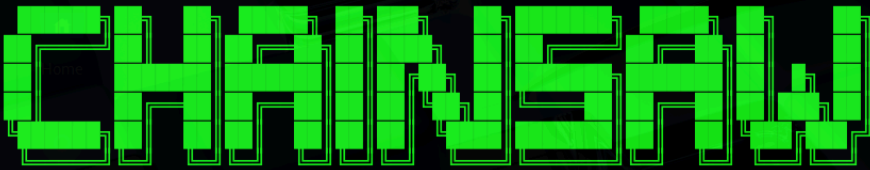
(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage]
$ ls
'Domain Controller'  Workstation

(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage]
$ cd 'Domain Controller/'

(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage/Domain Controller]
$ ls
SECURITY-DC.evtx
```

After downloading the file from HacktheBox, then unzip it and then we find the Triage folder. And inside the Triage folder, there is the Domain Controller folder and Workstation. Then inside the Domain Controller folder have the security logs.

```
(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage/Domain Controller]
$ chainsaw dump SECURITY-DC.evtx --json > security.json
```

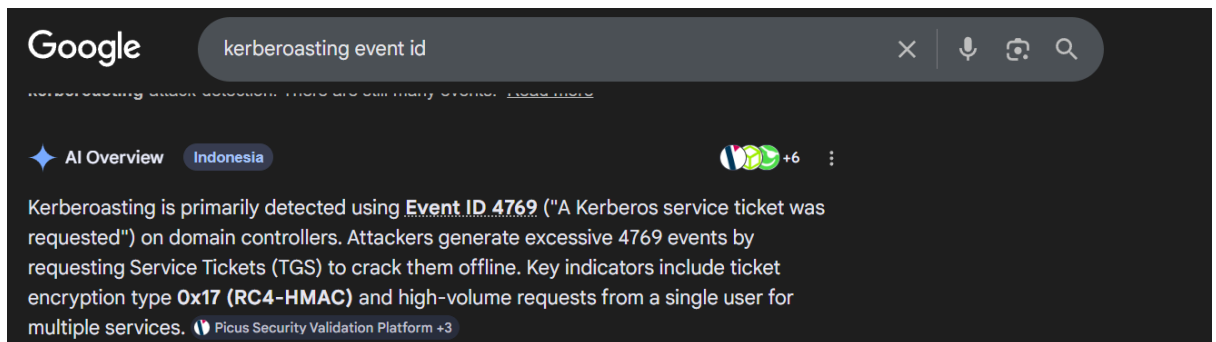


By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

```
[+] Dumping the contents of forensic artefacts from: SECURITY-DC.evtx (extensions: *)
[+] Loaded 1 forensic artefacts (1.1 MiB)
[+] Done
```

Then I Chainsaw dumped the security domain controller evtx to security.json for better visibility.

1. Analyzing Domain Controller Security Logs, can you confirm the UTC date & time when the kerberoasting activity occurred?



I searched in Google for Kerberoasting Event ID, and it is 4769.

```
System : {
  "Provider_attributes": {
    "Name": "Microsoft-Windows-Security-Auditing",
    "Guid": "54849625-5478-4994-A5BA-3E3B0328C30D"
  },
  "EventID": 4769,
  "Version": 0,
  "Level": 0,
  "Task": 14337,
  "Opcode": 0,
  "Keywords": "0x8020000000000000",
  "TimeCreated_attributes": {
    "SystemTime": "2024-05-21T03:18:09.459682Z"
  },
  "EventRecordID": 6672,
  "Correlation": null,
  "Execution_attributes": {
    "ProcessID": 748,
    "ThreadID": 5376
  },
  "Channel": "Security",
  "Computer": "DC01.forela.local",
  "Security": null
},
EventData: {
  "TargetUserName": "alonzo.spire@FORELA.LOCAL",
  "TargetDomainName": "FORELA.LOCAL",
  "ServiceName": "MSSQLService",
  "ServiceSid": "S-1-5-21-3239415629-1862073780-2394361899-1105",
  "TicketOptions": "0x40800000",
  "TicketEncryptionType": "0x17",
}
```

I found the event ID of 4769 and the ticket encryption type to be 0x17, therefore I get the system time and that was the answer, **2024-05-21 03:18:09**.

2. What is the Service Name that was targeted?

```
"Version": 0,  
"Level": 0,  
"Task": 14337,  
"Opcode": 0,  
"Keywords": "0x8020000000000000",  
"TimeCreated_attributes": {  
  "SystemTime": "2024-05-21T03:18:09.459682Z"  
},  
"EventRecordID": 6672,  
"Correlation": null,  
"Execution_attributes": {  
  "ProcessID": 748,  
  "ThreadID": 5376  
},  
"Channel": "Security",  
"Computer": "DC01.forela.local",  
"Security": null  
},  
"EventData": {  
  "TargetUserName": "alonzo.spire@FORELA.LOCAL",  
  "TargetDomainName": "FORELA.LOCAL",  
  "ServiceName": "MSSQLService",  
  "ServiceSid": "S-1-5-21-3239415629-1862073780-2394361899-1105",  
  "TicketOptions": "0x40800000",  
  "TicketEncryptionType": "0x17",  
  "IpAddress": "::ffff:172.17.79.129",  
  "IpPort": "58107",  
  "Status": "0x0",  
  "LogonGuid": "59F3B9B1-65ED-A449-5AC0-8EA1F68478EE",  
  "TransmittedServices": "-"  
}
```

Then I get the answer again from the same event as the previous questions and since it asked for the system name, there is also an attribute called system name so I just took that one which is **MSSQLService**.

3. It is really important to identify the Workstation from which this activity occurred. What is the IP Address of the workstation?

```
},
"EventID": 4769,
"Version": 0,
"Level": 0,
"Task": 14337,
"Opcode": 0,
"Keywords": "0x8020000000000000",
"TimeCreated_attributes": {
  "SystemTime": "2024-05-21T03:18:09.459682Z"
},
"EventRecordID": 6672,
"Correlation": null,
"Execution_attributes": {
  "ProcessID": 748,
  "ThreadID": 5376
},
"Channel": "Security",
"Computer": "DC01.forela.local",
"Security": null
},
"EventData": {
  "TargetUserName": "alonzo.spire@FORELA.LOCAL",
  "TargetDomainName": "FORELA.LOCAL",
  "ServiceName": "MSSQLService",
  "ServiceSid": "S-1-5-21-3239415629-1862073780-2394361899-1105",
  "TicketOptions": "0x40800000",
  "TicketEncryptionType": "0x17",
  "IpAddress": " ::ffff:172.17.79.129",
  "IpPort": "58107",
  "Status": "0x0",
```

Still in the same event, here there is an IP of the client getting attacked which is **172.16.79.129**

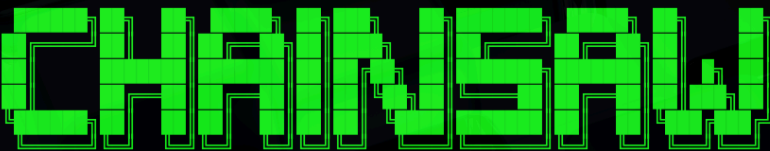
4. Now that we have identified the workstation, a triage including PowerShell logs and Prefetch files are provided to you for some deeper insights so we can understand how this activity occurred on the endpoint. What is the name of the file used to Enumerate Active directory objects and possibly

find Kerberoastable accounts in the network?

```
(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage]
$ cd Workstation

(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage/Workstation]
$ ls
2024-05-21T033012_triage_asset Powershell-Operational.evtx

(rafael@raf)-[~/HTB_SHERLOCKS/Campfire1/Triage/Workstation]
$ chainsaw dump Powershell-Operational.evtx --json > powershell.json
```



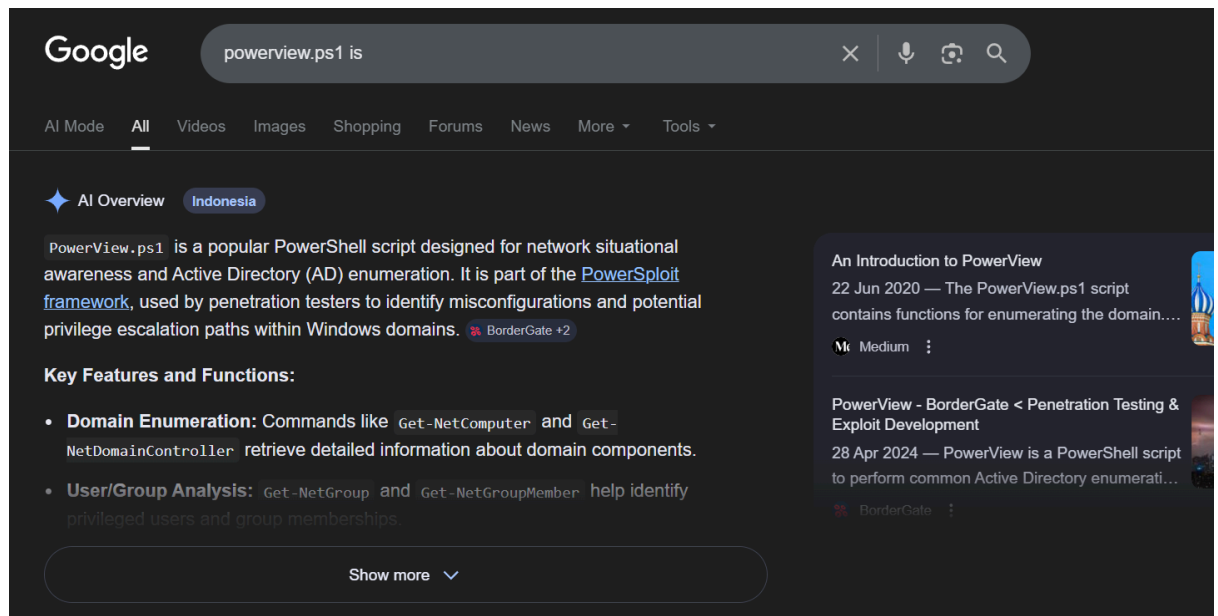
By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

```
[+] Dumping the contents of forensic artefacts from: Powershell-Operational.evtx (extensions: *)
[+] Loaded 1 forensic artefacts (2.1 MiB)
[+] Done
```

For this question, I went into the other folder which is Workstation and I turned the Powershell-Operational.evtx into powershell.json using Chainsaw dump.

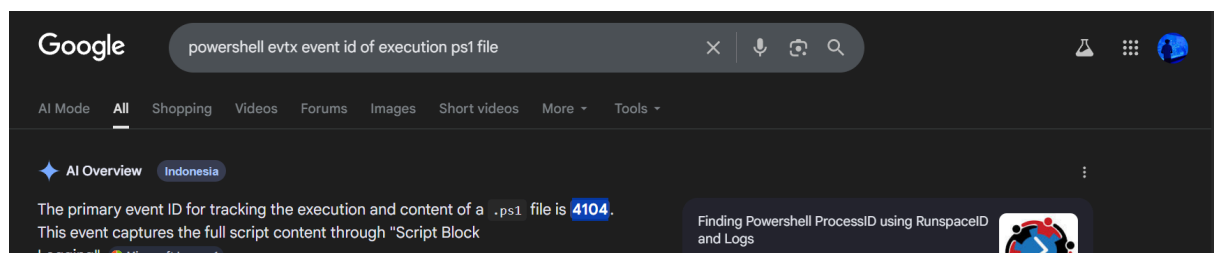
```
{
  "EventData": {
    "MessageNumber": 1,
    "MessageTotal": 1,
    "ScriptBlockText": "{\n                                # convert/process the LDAP fields f\n    $_.Properties\n    }",
    "ScriptBlockId": "5196972d-d363-4209-b1b9-13b1cc1584cb",
    "Path": "C:\\Users\\alonzo.spire\\Downloads\\powerview.ps1"
  },
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-PowerShell",
        "Guid": "A0C1853B-5C40-4B15-8766-3CF1C58F985A"
      },
      "EventID": 4104,
      "Version": 1,
      "Level": 3,
      "Task": 2,
      "Opcode": 15,
      "Keywords": "0x0",
      "TimeCreated_attributes": {
        "SystemTime": "2024-05-21T03:17:25.370783Z"
      },
      "EventRecordID": 166,
      "Correlation_attributes": {
```

I tried to cat the powershell.json to see what we have. Then I started to see a lot of these powerview.ps1 files. So I decided to Google it.



And it is used for Active Directory enumeration, and so the answer is **powerview.ps1**.

5. When was this script executed? (UTC)



I Googled first what is the Event ID of executing a ps1 file in Powershell and it turns out to be 4104

```
"EventID": 4104,
"Version": 1,
"Level": 3,
"Task": 2,
"Opcode": 15,
"Keywords": "0x0",
"TimeCreated_attributes": {
  "SystemTime": "2024-05-21T03:16:32.753612Z"
},
"EventRecordID": 160,
"Correlation_attributes": {
  "ActivityID": "B6DF577B-AB2B-0007-456C-DFB62BABDA01"
},
"Execution_attributes": {
  "ProcessID": 9200,
  "ThreadID": 9056
},
"Channel": "Microsoft-Windows-PowerShell/Operational",
"Computer": "Forela-Wkstn001.forela.local",
"Security_attributes": {
  "UserID": "S-1-5-21-3239415629-1862073780-2394361899-1104"
},
"EventData": {
  "MessageNumber": 1,
  "MessageTotal": 1,
  "ScriptBlockText": "{{$ _ -is [Reflection.Emit.ModuleBuilder]} -or ($ _ -is [Reflection.Assembly])}",
  "ScriptBlockId": "b8ad3c2e-2c9e-4649-b0d9-ecb91c312d29",
  "Path": "C:\\Users\\alonzo.spire\\Downloads\\powerview.ps1"
}
```

And then i saw this and the answer is **2024-05-21 03:16:32**.

6. What is the full path of the tool used to perform the actual kerberoasting attack?

The screenshot shows a Google search for "tools for kerberoasting attacks". The search results include a link to "Kerberoasting Attack - Detection and Prevention Strategies" by Netwrix, a link to "Steal or Forge Kerberos Tickets: Kerberoasting" by MITRE ATT&CK, and a link to "What is a Kerberoasting Attack? - CrowdStrike". The knowledge panel on the right provides a detailed breakdown of each step involved in performing a Kerberoasting attack, including reconnaissance, enumeration, and ticket extraction.

I Googled what are the tools used for kerberoasting attacks and I found it to be Impacket, Rubeus, Powersploit but I found no results.

Then I searched for rubeus and found something.

```
(venv)-(rafael@raf)-[~/../Workstation/2024-05-21T033012_triage_asset/C/Windows]
$ ls prefetch | grep -i rubeus
RUBEUS.EXE-5873E24B.pf
```

Then ChatGPT made me a script, to search the prefetch folder for any related files.

```
(venv)-[rafael@raf] - [~/.../Workstation/2024-05-21T033012_triage_asset/C/Windows]
$ python - << 'EOF'
import pyscca
pf = pyscca.open("prefetch/RUBEUS.EXE-5873E24B.pf")
print("\n".join([pf.get_filename(i) for i in range(pf.number_of_filenames)]))
EOF
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\NTDLL.DLL
\\VOLUME{01d951602330db46-52233816}\\USERS\\ALONZO.SPIRE\\DOWNLOADS\\RUBEUS.EXE
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\MSCOREE.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\KERNEL32.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\KERNELBASE.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\LOCALE.NLS
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\APPHelp.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\APPATCH\\SYSMAIN.SDB
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\ADVAPI32.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\MSVCRT.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\SECHOST.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\RPCRT4.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\BCRYPT.DLL
\\VOLUME{01d951602330db46-52233816}\\WINDOWS\\SYSTEM32\\URT\\GDI\\GDI32.dll
```

And here we found the folder which is

C:\Users\Alonzo.spire\Downloads\Rubeus.exe.

7. When was the tool executed to dump credentials? (UTC)

```
(venv)-(rafael@raf)-[~/../Workstation/2024-05-21T033012_triage_asset/C/Windows]
$ python - << 'EOF'
import pyscca

pf = pyscca.open("prefetch/RUBEUS.EXE-5873E24B.pf")

times = []

# Try up to 8 timestamps (Windows stores max 8)
for i in range(8):
    try:
        t = pf.get_last_run_time(i)
        if t:
            times.append(t)
    except:
        pass

for t in sorted(times, reverse=True):
    print(t)
EOF
2024-05-21 03:18:08.972822
1601-01-01 00:00:00
1601-01-01 00:00:00
1601-01-01 00:00:00
1601-01-01 00:00:00
```

Then, I used a script to find the UTC time for the file from the previous question, and then since the question asked for when the tool was executed, so I grabbed the first one, so that would be the answer which is **2024-05-21 03:18:08**.



You have solved Campfire-1!

Congratulations  **rafaelangelo** best of luck in capturing flags ahead!

#4658

Sherlock Rank

16 Apr 2026

Solve Date

Retired

Sherlock State

Ok

Share